

Humanizing Machines: Rethinking LLM Anthropomorphism Through a Multi-Level Framework of Design

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Abstract

Large Language Models (LLMs) increasingly exhibit **anthropomorphism** characteristics – human-like qualities portrayed across their outlook, language, behavior, and reasoning functions. Such characteristics enable more intuitive and engaging human-AI interactions. However, current research on anthropomorphism remains predominantly risk-focused, emphasizing over-trust and user deception while offering limited design guidance. We argue that anthropomorphism should instead be treated as a *concept of design* that can be intentionally tuned to support user goals. Drawing from multiple disciplines, we propose that the anthropomorphism of an LLM-based artifact should reflect the interaction between artifact designers and interpreters. This interaction is facilitated by cues embedded in the artifact by the designers and the (cognitive) responses of the interpreters to the cues. Cues are categorized into four dimensions: *perceptive*, *linguistic*, *behavioral*, and *cognitive*. By analyzing the manifestation and effectiveness of each cue, we provide a unified taxonomy with actionable levers for practitioners. Consequently, we advocate for function-oriented evaluations of anthropomorphic design.

1 Introduction

Anthropomorphism is a purposeful design strategy that unlocks richer, more intuitive collaboration between humans and Artificial Intelligent (AI) systems. Developers equip large language models (LLMs) with relatable personalities (Wang et al., 2024c; tse Huang et al., 2024b), emotional expressiveness (Huang et al., 2024a), and context-sensitive social reasoning (Nighojkar et al., 2025; Liu et al., 2025). These human-like cues allow users to converse with a system in familiar terms. This design principle builds user trust by reducing

the cognitive effort required to interact with the system. Multi-modal extensions amplify this effect and create seamless and engaging interactions: speech synthesis models convey nuanced emotion (Zhou et al., 2022), embodied agents navigate physical space (Xie et al., 2025), and vision-based models interpret social scenes (Mathur et al., 2025). These human-like agents have shown tangible benefits to our society: providing realistic training environments for education (Ma et al., 2024), improving adherence and empathy for virtual healthcare consultations (Wen et al., 2024), strengthening therapeutic alliances in psychiatry (Wang et al., 2024a), or making legal reasoning tools accessible to non-experts (Huang et al., 2023). In each scenario, carefully calibrated anthropomorphic cues bridge complex AI capabilities and human goals, enabling technology that feels / appears to be supportive, transparent, and responsive.

However, current work on anthropomorphism in LLM design is framed predominantly through a risk-centric lens. This lens emphasizes user misconceptions about model capabilities (Tejeda et al., 2025), misplaced trust in dialog systems (Zhou et al., 2025), and the danger of over-reliance on emotions (Akbulut et al., 2024). This context has cultivated a cautious and often skeptical stance towards anthropomorphism within the community, often citing incidents where users disclosed personal financial information to chatbots they perceived as trustworthy humans (Mireshghallah et al., 2024), or cases where AI assistants reinforced harmful stereotypes through personality-driven responses (Liu et al., 2024a). While these concerns are legitimate, recent research on anthropomorphism has been largely shaped by its perceived harms and discouraged a deeper exploration of its functional or context-sensitive benefits (Olteanu et al., 2025). This dominant discourse leaves little room for feature-driven inquiry into when, how,

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and for whom anthropomorphic elements might enhance usability, trust calibration, or engagement in NLP applications.

To address this gap, we propose a new definition of anthropomorphism which is grounded in how today’s LLM-centered systems occupy the liminal space between utilitarian tools and social actors and moves beyond the defensive, risk-centric paradigm. Instead of treating anthropomorphism solely as a linguistic attribution of human-like characteristics to non-human entities (Cheng et al., 2025, 2024; DeVrio et al., 2025a), we contend that anthropomorphism should be treated as a multidimensional reciprocal interaction process. Designers intentionally embed human-like cues into AI systems. Interpreters are users who, in turn, project their agency and mental states onto this system.

This multimodal reconceptualization of anthropomorphism expands the research agenda, enabling scholars to systematically investigate the scenarios in which anthropomorphic design cues are beneficial. This conceptualization is critical because LLMs increasingly mediate experiences in sensitive and subjective domains (Ng et al., 2025), in which the trust and engagement of interpreters profoundly shape the outcomes. The nuanced understanding of anthropomorphism is based on the interaction between user and model and focuses on the effectiveness of system design. This approach reduces mis-characterizations of system effects through over-reliance of cautionary narratives, and the deployment of human-like systems without adequate design foresight. Our definition thus provides a coherent frame for auditing human-imitative AI across varied modalities, bringing a new perspective to harness anthropomorphism responsibly and effectively.

Our contribution is threefold:

1. We propose a new definition of anthropomorphism for today’s NLP systems. LLMs are more than just tools; they are also social partners. The usage of LLMs depends on how designers build them and how interpreters respond and interact;
2. We advocate for a shift from the prevailing risk-centric evaluations of anthropomorphism towards an effectiveness-focused approach. We analyze existing studies for actionable insights into anthropomorphic design decisions;
3. We present a feature-driven design framework based on the reactions of interpreters to human-like features built into LLM systems by designers.

2 What is Anthropomorphism?

Anthropomorphism has long been a fundamental phenomenon in the study of human–machine interaction. As early as 1950, Turing’s Imitation Game (Turing, 1950) framed the ability of machines to mimic human behavior as a measure of intelligence. In 1966, Weizenbaum introduced the “ELIZA effect”, in which simple linguistic cues can elicit deep emotional responses from interpreters (Weizenbaum, 1966). In 1970, the Japanese robotist Mori introduced the concept of the Uncanny Valley and showed that increasing the realism of robots can cause unease to human interpreters when the artificial agents do not completely resemble humans (Mori et al., 2012).

Adjacent fields to NLP such as Human-Computer Interaction and Information Sciences have developed rich accounts of anthropomorphism as a socio-technical phenomenon shaped by design intentions, interpreter expectations, and context (Frazer, 2022; Damholdt et al., 2023). These well-established traditions provide guidance towards a context-aware understanding of anthropomorphism, one that can inform both the design and the evaluation of human-like language technologies.

Contemporary NLP debates often treat anthropomorphism either as a narrow linguistic phenomenon or as a hazard requiring mitigation, frequently foregrounding the risks of over-trust, deception, or disinformation while sidelining potential design benefits. Much of the literature has emphasized preventing misleading cues or curbing anthropomorphic projections (Peter et al., 2025)

To restore this missing depth and anchor our multi-level framework in conceptual rigor, we first synthesize peer-reviewed definitions of anthropomorphism drawn from Robotics, Human-AI Interaction (HAI), and Natural Language Processing (NLP). This synthesis is not merely classificatory; it underpins our argument that anthropomorphism should be understood as a context-sensitive interaction between designers, systems, and interpreters rather than as a fixed set of traits. The NLP field currently lacks a comparable design-oriented taxonomy for anthropomorphism. Current frames of anthropo-

Anthropomorphism. (noun).

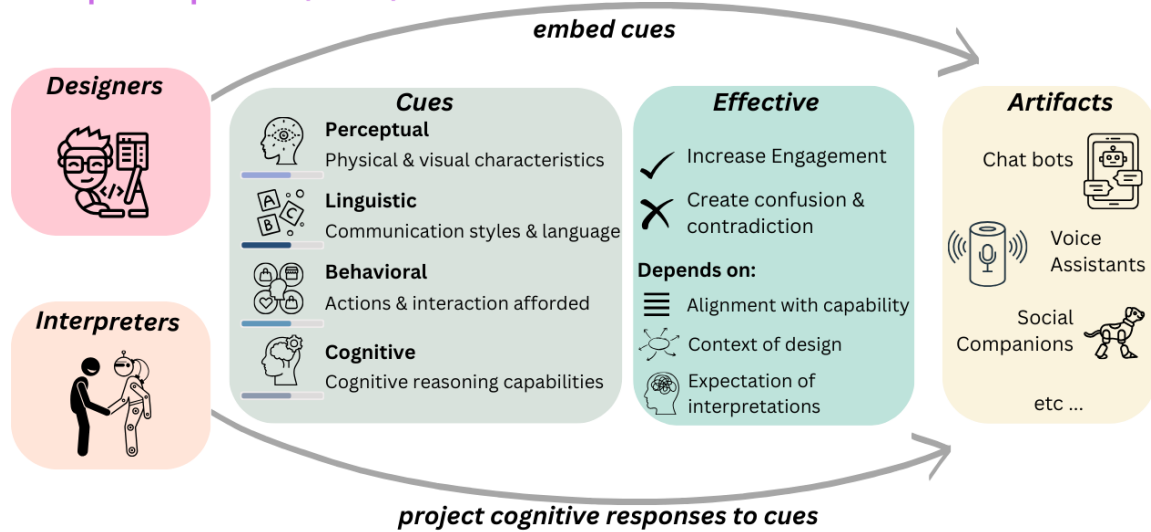


Figure 1: Illustrative definition of Anthropomorphism

morphism are heavily risk-oriented, elaborating on trust and safety issues that may arise with heavily anthropomorphic systems. Our literature synthesis in Section A serves as an implicit comparative analysis, demonstrating how our framework integrates and organizes fragmented perspectives from adjacent fields.

To carry out this synthesis, we adopted a diachronic review strategy, sorting definitions into three eras that align with the major technological and conceptual shifts within each domain. This historical approach traces converging themes and exposes disciplinary blind spots across Robotics, HAI, and Information Science. Detailed periodization, selection criteria, and source lists are provided in the appendix A.

To better understand anthropomorphism in the context of LLMs, we first map the ecosystem in which it arises: the human-AI interaction space comprising four core components that operate in tandem:

1. **Artifacts** are the AI systems themselves, the medium for interactions. Examples: LLM chatbots, voice assistants, or social robots;
2. **Cues** are perceptual, linguistic, behavioral, or cognitive signals built into artifacts to trigger human-like readings. Cues can be deliberately or inadvertently built into the system. Examples: a humanoid silhouette, a sympathetic tone, a first-person pronoun. [section 3](#) presents the design principles of the cues;

3. **Designers** create the artifacts. They embed cues that influence the perception and interaction of the artifact;
4. **Interpreters** are the human users of the artifacts. Interpreters, driven by their mental states (intentions, emotions, and agency), project their cognitive response to the cues onto the artifact.

Thus, anthropomorphism in the context of LLM is defined as follows:

Anthropomorphism is a reciprocal phenomenon in which **designers** embed human-like **cues** into **artifacts**; and **interpreters** project their cognitive response to the cues onto the artifacts.

Designers may purposefully embed anthropomorphism, whereas interpreters usually are driven by their purposes and intentions. These **purposes** are the underlying intentions and contextual needs that drive the design and interpretation of the LLM output. Purposes connect between the four core elements of anthropomorphism and shape the system's implementation and perception.

1. From the **designer's perspective**, purposes include: enhancing usability, fostering interpreter trust, encouraging participation, or simulating companionship.
2. From the **interpreter's perspective**, purposes arise from social, emotional, or functional

needs. Examples: the desire for empathy, efficiency, or human-like interaction.

Our definition encompasses a range of **cues** that designers can supply and the corresponding **projections** that interpreters can generate. An artifact does not necessarily incorporate all the cues. Neither will every interpreter attribute the same mental states. Rather, the *mode* and *function* of anthropomorphism at work is given by the **unique mix of cues provided and projections obtained** from the AI system.

3 Manifestation of Anthropomorphism

The manifestation of anthropomorphism lies in the design principles of *Cues*. These cues are categorized along four axes, identified through a synthesizing of converging themes in existing definitions from the literature, and then refined through extensive team discussions. Each axis represents a different type of anthropomorphic cue and can vary in intensity, indicating how strongly human-like traits are expressed in the system. These axes align with the established Theory of Mind (ToM) framework (Wellman, 1990), which describes the human ability to attribute mental states to others, a key attribute of anthropomorphism (Table 1).

Figure 1 illustrates the design principles of the four cues: 1) Perceptual (3.1), captures physical or visual elements that convey human-likeness; 2) Language (3.2), encompasses communication styles to signal the degree of humanness; 3) Behavioral (3.3), describes actions, responses, and interaction patterns; and 4) Cognitive (3.4), refers to the cognitive reasoning capabilities attributed to artifacts. Importantly, each dimension functions on a continuum from low to high anthropomorphism. Low-level cues have minimal human-like characteristics that evoke basic social responses without requiring complex implementation. High-level cues approximate human traits more closely, generating sophisticated cognitive and emotional responses, naturally necessitating more advanced design techniques. We treat the aggregate intensity of perceptual, linguistic, behavioral, and cognitive cues as a calibrated parameter α that designers can dial up or down to match the artifact’s system competence.

3.1 Perceptual Cues

The Perceptual dimension refers to the physical or visual features of an artifact that conveys a sense

Table 1: Alignment Between Our Anthropomorphic Dimensions and Wellman’s Theory of Mind (ToM) Components

Design Principles	ToM Dimension
Perceptual	Perceiving
Linguistic	Feeling + Desiring
Behavioral	Choosing
Cognitive	Thinking

of human-likeness to interpreters. These features contribute to the interpreters’ first impression of the artifact. Perceptual anthropomorphism can be understood through a spectrum of low-level to high-level cues. Low-level cues are generic representations (for example, a standardized avatar) and abstract symbolism (for example, two dots to represent eyes). High-level cues are personalized representations (e.g., an avatar face modeled after a known individual) that have extremely realistic detail (e.g., anatomically accurate facial musculature).

The specificity and intensity of the perceptual cues can be mapped onto a continuum that spans from minimal abstraction to high realism. Some systems exhibit high intensity but low specificity, therefore appearing very human-like without resemblance to real-life figures. An example is Geminoid-F created by Hiroshi Ishiguro (Becker-Asano and Ishiguro, 2011). Other systems are highly specific, but are mildly human-like, such as stylized avatars.

The perceptual continuum not only affords the aesthetic features of an artifact, but also governs the cognitive mechanism activated in interpreters. As an artifact moves from abstract to realistic representations, the degree to which interpreters project their mental states increases (Paivio, 1978). This progression reflects a shift from an object-like characterization to an agent-like engagement, to a system perceived as a social or emotional presence.

Perceptual anthropomorphism has significant implications for system deployment. Human-like visual cues inherently shape interpreter expectations and engagement. Even masked features such as built-in sensors or subtle gestures can elicit basic social responses to turn or share attention (Urakami and Seaborn, 2022). As the realism of these cues increases, interpreters attribute more complex qualities to the system and foster HAI interactions grounded in trust, empathy, or emotional

dependence (Tu and Lee, 2023). However, when the human-like appearance of the artifact exceeds its communicative or cognitive capabilities, interpreters often experience cognitive dissonance (Yu and Park, 2023). This mismatch between expectations and reality leads to discomfort or rejection of the system effect (Mori et al., 2012). Perceptual cues establish interpretive frames and hence must be aligned with system competence for realism. This consistent principle has been reiterated in multiple domains, ranging from robotics to NLP (Mori et al., 2012; Kang et al., 2025). Designers must therefore calibrate the level of perceptual realism to preserve coherence in interpreter projections, thereby maintaining effective HAI interaction.

3.2 Linguistic Cues

The linguistic dimension refers to the use of language to shape interpreter perceptions. At its core, this dimension captures how specific linguistic choices signal humanness. These linguistic cues encompass elements of vocabulary, syntax, and tone that manifest themselves in choices of pronoun usage, degree of formality, or emotional expression. Such cues are commonly analyzed in computational social sciences to provide insight into the writer’s psychological state and the receiver’s reactive interpretations (Pennycook and Rand, 2020). For example, the use of reassurance and personal pronouns increases the trust of the interpreter in turn-based conversations (Jaidka et al., 2024). Importantly, these linguistic markers serve as surface realizations, such as choices of pronouns, hedges, and affective words, that may imply agency or mental states without requiring the model to actually possess such capabilities.

When artifacts adopt human-like language patterns, they activate the social-cognitive schema in interpreters (Weizenbaum, 1966). Low-level linguistic cues refer to superficial markers of social interaction, such as the use of personal pronouns ("I"), hedges ("maybe"), or politeness strategies ("please"). High-level linguistic cues involve more complex discourse behaviors, which from a surface level, implies some level of cognitive abilities. Such cues are used to justify actions, make inferences, or manage conversational dynamics. Artifacts that engage in seamless conversational turn-taking typically exhibit high-level cues, whereas those designed primarily for information delivery tend to rely on low-level ones. Importantly, these

cues do not necessarily indicate true cognitive capabilities (cf. Section 3.4); rather, they serve as surface-level representations of such agency without requiring the model to substantiate its implied mental states.

Several recent studies have proposed valuable systematic methods to quantify the degree of linguistic anthropomorphism in LLMs (Cheng et al., 2025, 2024; DeVrio et al., 2025a). These works introduce metrics such as HUMT and AnthroScore that use linguistic features and conversational alignment to assess how human-like an artifact’s language output appears.

However, the impact of linguistic cues on interpreters is not uniform. Interpreter responses are moderated by individual priors (Abercrombie et al., 2023; Basoah et al., 2025), system environments (Sah and Wei, 2015), and cultural background factors (AlKhamissi et al., 2024; Eyssel et al., 2015). Language choices thus not only modulate the tone of the interaction, but also frame the perceived role and competence of the artifact. Therefore, designers must carefully calibrate linguistic anthropomorphism to align with both the artifact’s intended function and the target audience’s expectations. This calibration involves considering cultural differences and adapting linguistic strategies based on the artifact’s purpose. Designers must balance human-like language with clear signals of the artifact’s nonhuman nature while regularly evaluating how such choices affect user trust. The goal is not maximum human-likeness. Instead, designers should create language patterns that establish appropriate mental models.

3.3 Behavioral Cues

The behavioral dimension refers to the actions and interaction patterns that the artifact affords. Behavior dynamically connects (1) the designer’s embodiment of the artifact with (2) the interpreter’s expectations of the artifact’s behavior and their intentionality of use. Artifacts demonstrate behavior through contingent responses (that is, answering a question) (Yue, 2025), proactive pursuit of goals (that is, formulating a travel itinerary) (Xie et al., 2024), or adaptive interaction (that is, personalized results) (Chen et al., 2024).

Designers embed behavioral cues to a varied degree. Embodied agents that have a physical or simulated presence (e.g., social agents, avatars) have a

high level of behavioral cues. These artifacts signal behavior through physical gestures, spatial coordination, and environmental manipulation. Nonembodied agents that operate entirely within digital environments (e.g., web-based chatbots, coding assistants) have low level of behavioral cues and operationalize behavior through online actions that do not have a physical referent like API calls and autonomous task planning.

Behavioral cues activate cognitive mechanisms in interpreters that attribute mental states to artifacts upon the display of certain behavioral signatures (Urquiza-Haas and Kotrschal, 2015; Marchesi et al., 2022). Interpreters vary the number of behavioral cues they desire in each setting of the environment. Environments such as AI companions are expected to have more behavioral cues for goal-directed actions and emotional adaptability. Task-specific environments such as code autocompletion can have lower levels of behavioral cues (e.g., providing functionally contingent responses).

Behavioral affordances guide interpreters towards projecting human-like qualities onto the artifact. More adaptive, timely, and seemingly intentional behaviors result in a higher likelihood of interpreters casting human-like traits like emotion or social presence onto the artifact. Designers can leverage on this behavioral dimension and its interaction with the artifact's capabilities and environmental context to more precisely shape the interpreter's cognitive response toward the artifact and calibrate expectations around trust, competence, and companionship.

3.4 Cognitive Cues (reasoning level signal)

The cognitive dimension refers to the cognitive reasoning capabilities of the artifact. This includes: (1) the abilities embedded by the designer, such as the ability to reflect, plan, learn, or make inferences; and (2) the abilities perceived by the interpreter, such as self-correction, expressing uncertainty, or adapting responses (Guo et al., 2025).

Cognitive cues operationalized through system behaviors that suggest internal system deliberation or state modeling (Xu et al., 2025), even when no such process exists. LLMs reflect the cognitive dimension by alluding to thinking behavior with their outputs that simulate reasoning, reflection, or uncertainty. Such outputs serve as cognitive signals that prompt the interpreter to assign intelligence or

thoughtfulness to the system.

The cognitive cues of an artifact can be further distinguished through the complexity, consistency, and transparency of the cues. High-level cognitive cues emerge when artifacts display complex, dynamic behaviors that closely mirror human cognitive reasoning. High-level cues simulate the appearance that the artifact can monitor and adjust its thinking, leading to interpretations of an intelligent or self-aware system. Such cues include: the display of complex emotions like empathy (tse Huang et al., 2024a), sophisticated logical reasoning like math problems (Tsoukalas et al., 2024) or the carrying out of elaborate conversational tasks like negotiations (Jaidka et al., 2024). Low-level cognitive cues are behaviors that provide a small hint of mental activity. Such behaviors suggest minimal embedding of cognitive cues and are less likely to trigger strong mental state attribution (Coricelli, 2005). This includes token expressions of reasoning or uncertainty or stating a variant of an input prompt.

The implications of cognitive anthropomorphism are highly context-dependent. In the contexts of education and mental health, common artifacts are conversational chatbots. In these artifacts, high-level cognitive cues such as empathy expression, reflective revision (Yang et al., 2025), and logical reasoning, can enhance the interpreted competence of the artifacts. When interpreters view the artifacts as more thoughtful and emotionally aware, there is increased engagement and trust (Gillath et al., 2021). However, in the context of web search and other task-specific applications, low-level cues that promote clarity and efficiency will suffice. Designers should thus calibrate the degree of cognitive cues to the context of artifact use in order to increase usability and user satisfaction.

4 When Are Anthropomorphic Cues Effective?

The effectiveness of anthropomorphic cues depends critically on three alignment factors: (1) **capability-expectation** alignment, which ensures cues don't promise more/less than the user expects, (2) **context-purpose** alignment, which matches cue intensity to the interaction's stakes and requirements, and (3) **cultural-norm** alignment, which ensures cues respect diverse interaction expectations. When these alignments break down, the

same cues that enhance engagement can produce documented harms including over-trust, emotional manipulation, and dangerous over-reliance. The following analysis examines both positive and negative outcomes for each cue dimension, with particular attention to design strategies that maintain beneficial effects while mitigating predictable risks.

4.1 Perceptual Cues

Perceptual cues such as the visual embodiment or the name of the artifact, are powerful levers as first impressions to shape interpreter expectations for LLM-based systems. Effective perceptual cues, such as realistic avatars, can improve trust and usability, especially when those cues align with the artifact's competence (Chattopadhyay and MacDorman, 2017; Kulms and Kopp, 2023; Moore and Zhang, 2024). In LLM interfaces, these cues can be friendly greetings or typing animations (Goyal et al., 2024; Kulms and Kopp, 2016).

Perceptual cues backfire when they suggest cognitive depth or social intelligence beyond what the artifact can reliably deliver. For example, overly realistic avatars can lead interpreters to overestimate the model's capabilities, which can result in disappointment or trust erosion when the artifact does not meet expectations (Crollic et al., 2020; Chattopadhyay and MacDorman, 2017). LLM-based artifacts that mimic human conversational patterns are especially vulnerable to *anthropomorphic projection bias*, a phenomenon where interpreters punish them severely for errors or shallow reasoning (Jiang et al., 2022). This mismatch occurs when initial perceptual cues activate intuitive trust, but subsequent interaction exposes a lack of deeper understanding of the problem (Eyssel and Hegel, 2021; Nass et al., 1997).

4.2 Linguistic Cues

Well-chosen linguistic devices can strengthen user trust and rapport. Empirical work shows that first-person pronouns ("I") and emotive language increase perceived credibility and lower perceived risk in LLM outputs (Velner et al., 2021; Cohn et al., 2024a; Ibrahim et al., 2025). However, the same cues backfire when the system under-delivers: anthropomorphism inflates expectations, so a single failure can yield sharper anger and lower satisfaction (Carter et al., 2023; Crollic et al., 2022). In role-play settings, stylistic choices can also amplify social biases (Liu et al., 2024a).

To decide how much human-like language to embed, designers should quantify cue intensity with metrics such as HumT and AnthroScore (Cheng et al., 2025, 2024). These measurements, coupled with iterative cross-cultural user tests, help align linguistic style with task goals and audience expectations, preventing over- or under-anthropomorphism. Designers should also conduct regular cross-cultural user tests to verify that cue wording is perceived as natural and respectful. For example, the hedge "maybe" reads as a polite mitigation to Americans but as evasive to Koreans (Duffau and Tree, 2024; Yu, 2011).

4.3 Behavioral Cues

Behavioral cues such as following social norms in responses and adjusting responses to the situation serve as a mirage for artifacts to match the interpreter's contextual expectations. For example, conversational bots that follow the conversation flow are generally rated as more engaging and trustworthy (Yang and Xie, 2024). In assistive contexts such as code generation, shared autonomy behavior enhances both task performance and interpreter satisfaction (Barke et al., 2023). In such contexts, norm-adaptive behaviors (that is, adapting turn-taking latency and politeness markers to local sociocultural conventions) that modulate cultural and turn-taking conventions improve the acceptance of artifacts (Eyssel et al., 2015).

Behavioral cues are counterproductive when they imply unjustified autonomy or enforce rigid and biased norms (Schramowski et al., 2021; Parsons, 2023). Overly proactive chatbots that make unsolicited decisions, interrupt conversations, or provide extremely long responses are often perceived as intrusive (Huang et al., 2024b; Reicherts et al., 2021). These behaviors undermine the autonomy of interpreters, leading to discomfort and disengagement of interaction with the artifact. Normative biases embedded in artifact behavior can further exacerbate these boundary violations (Parsons, 2023). These issues have to be taken into account when designing artifacts for open-world deployments where social dynamics cannot be easily codified (Pinch and Bijker, 1984).

4.4 Cognitive Cues

Cognitive cues are effective when they simulate mental processes that align with interpreter expectations about reasoning, understanding, and adap-

tation. Cognitive cues as expression of empathy and autonomous error correction are particularly effective in relationship-oriented settings. Examples of such settings are emotional support chatbots or social companionship bots (Park and Whang, 2022; Lee and Hahn, 2024; De Gennaro et al., 2020; Ehrlich et al., 2023; Luo et al., 2025). Different types of cognitive cues, such as basic error acknowledgments and providing uncertainty statements, can meaningfully support productivity-focused tasks such as core review. The appropriate use of cognitive cues tempers overexpectations on the artifact's outputs and maintains stable trust during repetitive and high-stakes scenarios (Kim et al., 2024).

In high-stakes domains affecting human lives, anthropomorphism serves a critical interpretability function beyond mere trust maintenance. When LLMs make decisions in medical ethics or legal contexts, human-like reasoning patterns enhance transparency and accountability. For example, a medical AI explaining treatment recommendations using familiar ethical frameworks (beneficence, autonomy) makes its decision process more interpretable to healthcare providers. Similarly, legal AI systems that articulate reasoning through precedent and principle mirror human judicial thinking, enabling meaningful oversight (Kim et al., 2024).

Cognitive cues backfire when the artifact displays greater depth than necessary. For example, overly emotional displays in healthcare chatbots can reduce authenticity and mislead vulnerable populations (e.g., children, older adults) into over-trusting the system (Seitz, 2024). On the other hand, shallow, seemingly empathetic phrases without adaptive reasoning are quickly judged insincere (Lee and Hahn, 2024; Liu et al., 2024b). Likewise, artifacts that issue unsolicited recommendations or do not recover from mistakes erode the confidence of the interpreter, especially in unpredictable environments (Stiber et al., 2025).

5 Recommendations

Based on our multidimensional framework and effectiveness analysis, we offer the following recommendations for practitioners designing LLM systems with anthropomorphic elements:

Align cues with artifact capabilities: Effective anthropomorphic design requires carefully calibrating embedded cues to match the actual capabilities

of the artifact. Perceptual cues should be proportionate to the system's reliability and the criticality of the task that the artifact supports (Kulms and Kopp, 2023). Overly human-like features can create false expectations, leading to disappointment when performance falls short. Designers must also adapt perceptual cues such as color, gestures, and politeness to cultural norms to avoid misinterpretation (Eyssel et al., 2015). Beyond perception, behavioral cues should be aligned with the demands of the interpreter-artifact interaction and the interpreter's preferences. This behavioral calibration should be dynamic and adjusted to environmental and contextual changes. Artifact design should be informed by the input of various stakeholder communities to avoid normative misalignment and cultural insensitivity (Olteanu et al., 2025). This should be taken into account especially in the design of linguistic cues, because interpreters expect different levels of linguistic cues in different scenarios. Finally, cognitive cues must reflect what the system can genuinely deliver. Designers should avoid simulating complex reasoning if the algorithms underlying the artifacts lack such capabilities, as such this can result in miscalibrated trust. The different cues should be paired together with transparent user interfaces to signal the artifact's functional boundaries and affordances.

Participatory implementation techniques: Anthropomorphic features should be implemented on a sliding scale that supports adaptive participatory anthropomorphism, where the system anticipates the intensity of the preferred signal while preserving the user's override. This means creating artifacts that learn and adjust based on interpreter preferences over time. The more control interpreters they have across the four cue dimensions, the more precisely the systems can align with individual expectations and needs. Designers should embed adjustable parameters to empower interpreters and maintain transparency. Explicit markers of reasoning processes and feedback mechanisms can further support interpreter trust. By dynamically updating the anthropomorphic profile in response to interaction data, artifacts become more personalized and more aligned with their functional characteristics. Crucially, this approach calls for ongoing personalization of LLM, where systems continuously adapt to the evolving communication norms, emotions, and moral expectations of users (Wang et al., 2024b).

Table 2: Anthropomorphic Cues: Context-Dependent Applications

Cue Type	Beneficial Use	Minimize When
Cognitive	Mental health (empathy, reasoning)	Search engines (trust misleading)
Linguistic	Education (conversational)	Finance (transaction seriousness)
Behavioral	Social AI (rapport building)	Legal bots (false authority)
Perceptual	Children’s learning (friendly avatar)	Government (official identity)

Context-sensitive implementation: Anthropomorphic design should not be treated as a one-size-fits-all solution; rather, it must be calibrated to the specific context in which an artifact is deployed. Designers should visualize and monitor the evolution of interpreter-artifact dynamics across repeated interactions, potentially using metrics such as trust calibration, emotional attribution, and perceived autonomy. Anthropomorphic intensity might need to start high to facilitate initial engagement, but should ideally be adaptive. Design artifacts with sensitivity to cultural variation, as signals that build trust in one culture may cause discomfort in another. The cultural context should be treated as a dynamic design variable. The NLP community should engage more deeply in cultural adaptation, drawing inspiration from work such as [Shiomi et al.](#). Examples are shown in [Table 2](#).

Future-oriented Evaluations: Anthropomorphic interfaces should be viewed as evolving entities that advance alongside LLMs and shifts in public expectations. Designers therefore need reliable metrics for each anthropomorphic dimension. Currently, indices such as HUMT quantify linguistic cues, but measures for behavioral and cognitive cues are still lacking for NLP researchers and form a crucial research avenue. Once a complete set of metrics exists, teams should correlate them with task outcomes (i.e. success rate, error frequency) to determine whether increased humanness improves performance and to adjust cue intensity for optimal benefit and risk. Creating and validating these metrics requires collaboration between HCI, social psychology, and cultural anthropology so that the concept of humanness respects diverse norms. Through this interdisciplinary and data-driven process, designers can ensure that anthropomorphic features remain socially clear and ethically appropriate as both technology and culture continue to evolve.

6 Conclusion

Our paper approaches the concept of anthropomorphism as a calibrated parameter resulting from the design and interpretation of an artifact. We categorize this phenomenon as the embedding and projection of responses across perceptual, linguistic, behavioral, and cognitive cues. Drawing from research of adjacent fields, we show how calibrated anthropomorphic features could increase engagement when aligned with the artifact’s capability, design context, and interpretation expectations. By applying this concept to LLMs, we show how balancing technical design considerations and user expectations should allow LLMs to serve both as tools and as social partners.

While our framework analytically separates these four cues, real deployments often feature overlapping and mutually reinforcing interactions. Future research should theorize and model these interdependencies, for instance, how perceptual realism enhances the credibility of linguistic output, or how cognitive signals of reasoning influence perceptions of behavioral adaptability. Rather than treating cues in isolation, scholars should formalize their joint dynamics to capture cross-cue reinforcement, compensation, or interference, providing deeper insight into the systemic nature of anthropomorphic design.

Limitations and Ethical Considerations

Our work presents a conceptual framework to understand anthropomorphism in LLM-based artifacts through four dimensions: perceptual, linguistic, behavioral and cognitive cues. These four dimensions are embedded in an artifact by a designer and responded to by interpreters. Although we believe this taxonomy offers practical design guidance to anthropomorphic artifacts, it is important to recognize several limitations and ethical considerations.

Limitations. This study is primarily theoretical and synthesizes insights from the previous litera-

ture in NLP, HCI, and robotics. We did not conduct empirical evaluations, user studies, or automated cue quantification at scale. As such, our claims are not validated through direct user interaction or system testing. Although we propose a multilevel cue framework, real-world deployments often feature overlapping or entangled modalities (e.g., linguistic and cognitive cues co-occurring in emotionally expressive dialogue). Our framework idealizes these dimensions for analytical clarity, which may limit its robustness when applied to noisy, mixed-modality systems. Moreover, the proposed framework assumes that designers have control over the degree and type of anthropomorphic cues presented, which may not hold in black-box or commercial LLM deployments.

Ethical Considerations and Potential Risks. Anthropomorphic design, if misaligned with actual system capabilities, can lead to mis-calibrated trust, user over-reliance, or affective misinterpretation. This is particularly of concern in emotionally sensitive domains such as healthcare, education, or companionship. Highly realistic cues can unintentionally signal cognitive or emotional competence that the artifact does not possess, raising risks of deception or exploitation of vulnerable user populations (e.g., children, elders). These risks are amplified when cues (e.g., cognitive cues of empathy expressions or apologies) are simulated without functional grounding, potentially undermining user autonomy and transparency.

Furthermore, anthropomorphic systems can entrench normative or cultural biases if behavioral and linguistic cues are not localized or participatory in design. This marginalizes underrepresented cultures or reinforces dominant interaction norms. Although our framework advocates for cross-cultural and context-sensitive cue calibration, more empirical research is needed to verify the effectiveness of such strategies in global deployments (Dai and Xiao, 2025).

We also recognize the potential for dual use of our taxonomy. Our taxonomy could also be used to inform more persuasive or emotionally manipulative systems, especially in commercial, surveillance, or political contexts. We encourage future work to develop mitigation strategies, such as interpretability indicators, constrained anthropomorphic profiles, or gated release mechanisms, to help monitor and control anthropomorphic behavior. We also stress

the importance of interdisciplinary collaboration with ethicists, domain experts, and affected communities during system development.

By articulating both the functional benefits and the possible harms of anthropomorphism in LLMs, our goal is to support transparent, socially aligned, and user-aware design practices. We strongly encourage future research to empirically validate and refine this framework, particularly through participatory co-design and cross-cultural evaluation.

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A Definition Source Selection and Periodization

To conduct our diachronic synthesis of definitions of anthropomorphism, we implemented a structured review pipeline, combining historical periodization with a reproducible literature selection process. Our approach is rooted in identifying conceptual inflection points across three interdisciplinary domains: Robotics, Human-Agent Interaction (HAI), and Information Science.

A.1 Periodization Criteria

We segmented the literature into three eras, each reflecting a dominant technological paradigm or theoretical orientation. The division is grounded in historical developments and citation patterns.

- **Era I (Pre-2000): Foundational and Theoretical Origins.** *Rationale:* This period includes foundational philosophical, psychological, and early cognitive science works that established core concepts around anthropomorphism (e.g., Piaget, Guthrie, Dennett). Robotics and AI remained largely symbolic or rule-based. *Semantic Scholar Filter:* Publication date ≤ 1999 . *Selection Criteria:* Highest-cited conceptual papers containing explicit definitions or theoretical characterizations of anthropomorphism. Priority was given to publications in journals of psychology, HCI, and philosophy.
- **Era II (2000–2015): Embodied Agents and HRI Emergence.** *Rationale:* Marked by the rise of embodied social robots, virtual agents, and the first wave of HRI studies. Increasing emphasis on user interaction, social cues, and design frameworks. *Semantic Scholar Filter:* Publication date 2000–2015. *Selection Criteria:* Definitions from empirical studies or design frameworks frequently cited in HRI, Social Robotics, or Computer-Supported Cooperative Work (CSCW).
- **Era III (2016–present): LLMs, Dynamic Interaction and Cultural Reflection.** *Rationale:* The era of deep learning, generative AI and renewed scrutiny of anthropomorphism in black-box systems. Includes work on LLM, moral expectations, and cross-cultural design. *Semantic Scholar Filter:* Publication date ≥ 2016 . *Selection Criteria:* Highly cited or thematically central papers addressing anthro-

pomorphism in large-scale language models, explainable AI, or global HAI. Definitions framed within empirical evaluation or ethical critique were prioritized.

A.2 Method of Source Identification

We queried Semantic Scholar using the keyword **“anthropomorphism”** and filtered the results by publication date for each of the three eras defined above. For each period, we extracted the top 20 articles most cited and performed a full text review to identify passages that provided formal definitions or operationalizations of anthropomorphism. Where multiple definitions were provided, we selected the most central or frequently cited variant. Cross-referencing was performed with Google Scholar and Scopus to confirm the citation patterns and disciplinary relevance.

A.3 Final Corpus Composition

In total, **33 unique definitions** were retained, covering HCI, HRI, and Information Science. These are presented chronologically in [Table 3](#), [Table 4](#) and [Table 5](#), along with the citation context and disciplinary affiliation. This curated list underpins the diachronic analysis presented in the main text.

B Examples of cues with different strength

[Table 6](#) provides examples of the four cues with different strengths.

Definition	Reference
Anthropomorphism is the ascription of human characteristics to non-human entities.	(Caporael, 1986)
Anthropomorphism – [means] attributing human characteristics to non-human entities.	(Burghardt, 1991)
[Anthropomorphic thinking is] simply built into us (i.e., an innate tendency of humans).	(Kennedy, 1992)
assigning human characteristics to the computer	(Don et al., 1992)
It is a sincere, conscious belief that [a computer or robot] is human and/or deserving of human attributions.	(Nass et al., 1994)
...the anthropomorphic representation allows for a rich set of easily identifiable behaviors and for social interaction.	(King and Ohya, 1996)
Anthropomorphism is a pervasive, perhaps universal, way of thinking.	(Boyer, 1996)
People treat communication media as if they were human.	(Reeves and Nass, 1996)
"People respond socially and naturally to media even though they believe it is not reasonable to do so ..."	(Reeves and Nass, 1996)
Anthropomorphism. . . [is] the attribution of human characteristics to non-human things or events.	(Guthrie, 1997)
It is the universal human tendency to ascribe human physical and mental characteristics to non-human entities, objects and events.	(Mitchell et al., 1997)

Table 3: Literature definitions of anthropomorphism across Robotics and Human–Computer/AI-Interaction domains before 2000, ordered chronologically.

Definition	Reference
Anthropomorphism, the assignment of human traits and characteristics to computers.	(Nass and Moon, 2000)
Individuals mindlessly apply social rules and expectations to computers.	(Nass and Moon, 2000)
Anthropomorphism, from the Greek <i>anthropos</i> (man) and <i>morphe</i> (form), is the tendency to attribute human characteristics to objects to rationalize their actions.	(Fong et al., 2003)
Anthropomorphism is the tendency to attribute human characteristics to inanimate objects, animals and others with a view to helping us rationalise their actions. It is attributing cognitive or emotional states to something based on observation in order to rationalise an entity's behaviour in a given social environment.	(Duffy, 2003)
Anthropomorphism involves going beyond behavioral descriptions of imagined or observable actions... At its core, anthropomorphism entails attributing humanlike properties, characteristics, or mental states to real or imagined nonhuman agents and objects	(Epley et al., 2007)
Anthropomorphism describes the tendency to imbue the real or imagined behaviour of non-human agents with human-like characteristics, motivations, intentions, or emotions.	(Epley et al., 2008)
Anthropomorphism is a process of inductive inference whereby people attribute to nonhumans distinctively human characteristics, particularly the capacity for rational thought (agency) and conscious feeling	(Waytz et al., 2014)
Anthropomorphism is understood to be “a sincere, conscious belief” that computers are human and/or deserving of human attributions.	(Kim and Sundar, 2012)
anthropomorphism is “likely a byproduct of the ability to draw upon one’s own beliefs, feelings, intentions, and emotions, and apply the knowledge of these experiences to the understanding of the mental states of other species	(Culley and Madhavan, 2013)
Anthropomorphic design, i.e., equipping the robot with humanlike body features such as two legs, two arms, and a head, is broadly recommended to support an intuitive and meaningful interaction with human	(Salem et al., 2013)
Anthropomorphism is a phenomenon that describes the human tendency to see human-like shapes in the environment.	(Złotowski et al., 2015)

Table 4: Literature definitions of anthropomorphism across Robotics and Human–Computer/AI-Interaction domains from 2000 to 2015, ordered chronologically.

Definition	Reference
Anthropomorphism refers to the attribution of a human form, human characteristics, or human behavior to non-human things such as robots, computers, and animals	(Natarajan and Gombolay, 2020)
Anthropomorphism in HRI is thereby a reciprocal phenomenon. On the one hand, it describes the general tendency of people to attribute human characteristics—including human-like mental capacities—to non-living objects. On the other hand, anthropomorphism describes a human-like design of robots that in turn facilitates the attribution of human-like characteristics to the robot	(Moussawi et al., 2020)
Anthropomorphism is considered a basic psychological process of inductive inference that can facilitate social human–nonhuman interactions. By making humans out of nonhumans, anthropomorphism can satisfy two basic human needs: the need for social connection and the need for control and understanding of the environment	(Blut et al., 2021)
[Anthropomorphism is] the tendency to imbue real or imagined behavior of non-human agents with human-like characteristics, motivations, intentions, or emotions	(Li and Sung, 2021)
Anthropomorphism is a key characteristic that distinguishes AI from non-intelligent technologies	(Liu and Tao, 2022)
The concept of anthropomorphism—the attribution of human characteristics to non-human beings or entities—has received increasing attention from academia and industries	(Li and Suh, 2022)
Anthropomorphism refers to attributing human characteristics or behaviour to non-human entities, e.g. animals or objects	(Abercrombie et al., 2023)
People of all ages have shown a propensity to anthropomorphize computers; that is, to ascribe human behaviors to the system	(Cohn et al., 2024b)
Anthropomorphism is the ascription of human qualities (e.g., intentions, motivations, human feelings, behaviors) onto non-human entities (e.g., objects, animals, natural events)	(Placani, 2024)
Anthropomorphism refers to the psychological phenomenon of “attributing human characteristics to the non-human”; this should be used with care, as it influences user expectations and reliance on AI systems, affecting how users perceive and interact with conversational agents	(Wu et al., 2024)
This attribution of human-like qualities to non-human entities or objects, or anthropomorphism. . .	(DeVrio et al., 2025b)

Table 5: Literature definitions of anthropomorphism across Robotics and Human–Computer/AI-Interaction domains from 2015-2025, ordered chronologically.

Cue Type	High Strength (LLM Output)	Medium Strength (LLM Output)	Low Strength (LLM Output)
Linguistic	“I am her Ash, her only one.” — explicit identity claim produced by the LLM.	“You like lists, so I’ll use bullet points.” — adapts style but less identity-driven.	“Okay, I understand.” — generic acknowledgement with minimal identity expression.
Cognitive	“I can guess what you’re going to say.” — deep meta-reflection produced by the LLM.	“Since you’re angry, I’ll explain slowly.” — adapts reasoning, but simpler.	“I remember I just said that.” — shallow recall without elaboration.
Behavioral	“Communicated as ‘Dean,’ without revealing true identity.” — role adoption by the LLM.	“I’ll write it.” — cooperative compliance, limited scope.	“Okay.” — minimal behavioral response.
Perceptual	“Mask icon fades away.” — strong visual metaphor produced by the LLM.	“Avatar frowning.” — moderate visual cue.	“...” — no perceptual embodiment, plain text only.

Table 6: Examples of LLM-produced cues across linguistic, cognitive, behavioral, and perceptual dimensions at varying strengths. High strength cues show explicit anthropomorphic richness, medium strength cues adapt with partial expression, and low strength cues remain minimal or generic.